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Report: Optimising NYC Taxi Operations

Include your visualisations, analysis, results, insights, and outcomes. Explain your methodology and approach to the tasks. Add your conclusions to the sections.

1. Data Preparation

1.1. Loading the dataset

The NYC Yellow Taxi trip records were provided as monthly parquet files. Since each file contains millions of records, loading all files together could cause memory issues. Therefore, the files were processed one by one using Pandas.

Each file was read separately and necessary datetime conversions were applied to prepare the data for further analysis.

1.1.1. Sample the data and combine the files

Due to the extremely large size of the dataset, analysing the complete data was computationally expensive. Hence, a sampling strategy was applied.

The sampling approach used was:

- The tpep_pickup_datetime column was converted to datetime format.
- Separate columns for pickup date and pickup hour were created.
- For each date and each hour, random sampling was performed.
- **5% of trips (frac = 0.05) were selected from each hour of every day.**
- A fixed random seed (random_state = 42) was used to ensure reproducibility.

This ensured that:

- All hours and days were equally represented.
- Peak and off-peak patterns were preserved.
- Seasonal and temporal variations were maintained.

All sampled monthly data were concatenated into a single DataFrame representing the yearly dataset.

Finally, the sampled data was saved in parquet/CSV format for faster access during analysis.

2. Data Cleaning

2.1. Fixing Columns

2.1.1. Fix the index

After loading the sampled dataset:

- The default index was reset.
- Unnecessary indices were removed.
- Column names were reviewed for consistency.

2.1.2. Combine the two airport_fee columns

Two airport fee columns were found, indicating a possible naming issue. These were merged into a single airport_fee column to avoid redundancy.

2.2. Handling Missing Values

2.2.1. Find the proportion of missing values in each column

The proportion of missing values was calculated for all columns. Most columns had negligible missing values.

2.2.2. Handling missing values in passenger_count

- Zero and missing values were replaced with the median value.
- This ensured realistic passenger counts.

2.2.3. Handle missing values in RatecodeID

Missing values were filled using the mode.

2.2.4. Impute NaN in congestion_surcharge

Missing values were replaced with 0.

After imputation, all columns were rechecked to ensure no missing values remained.

2.3. Handling Outliers and Standardising Values

2.3.1. Check outliers in payment type, trip distance and tip amount columns

Monetary columns such as fare_amount, tip_amount, tolls_amount, and total_amount were examined.

Rows with negative values in these columns were removed since negative fares are not realistic.

Outlier analysis using describe() revealed abnormal values such as:

- Extremely high trip distances
- Very high fares for short distances
- Invalid payment types

The following actions were taken:

- Removed trips with passenger_count > 6
- Removed trips with trip_distance > 250 miles
- Removed records with payment_type = 0
- Removed trips with near-zero distance but very high fare

These values were likely caused by recording errors and were dropped

3. Exploratory Data Analysis

3.1. General EDA: Finding Patterns and Trends

3.1.1. Classify variables into categorical and numerical

Variables were categorised as:

1. Numerical Variables

- passenger_count
- trip_distance
- fare_amount
- tip_amount
- total_amount
- tolls_amount

2. Categorical Variables

- VendorID
- RatecodeID
- payment_type
- PULocationID
- DOLocationID
- store_and_fwd_flag

3. Temporal Variables
    - tpep_pickup_datetime
    - tpep_dropoff_datetime
    - pickup_hour
    - trip_duration

All monetary variables were treated as numerical.

3.1.2. Analyse the distribution of taxi pickups by hours, days of the week, and months

Pickup distributions were analysed across:

- Hours
- Days of week
- Months

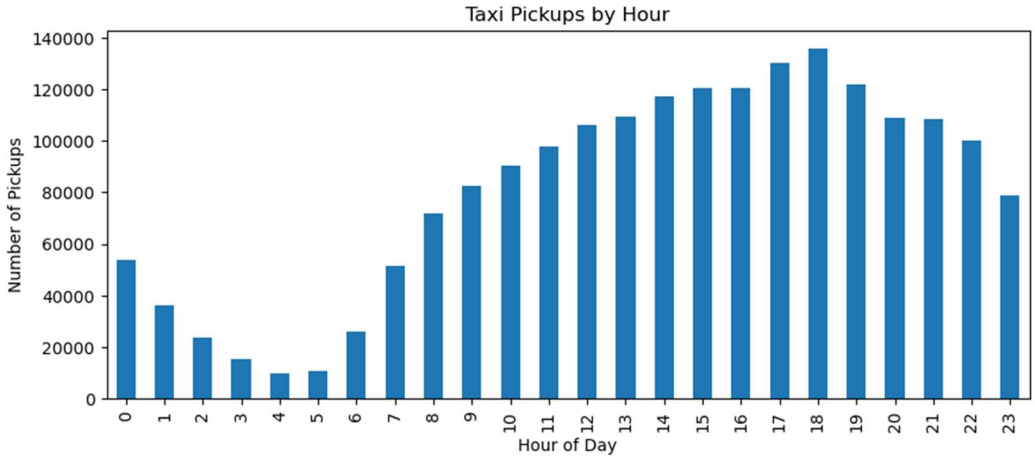


Fig.: Taxi pickups by hour

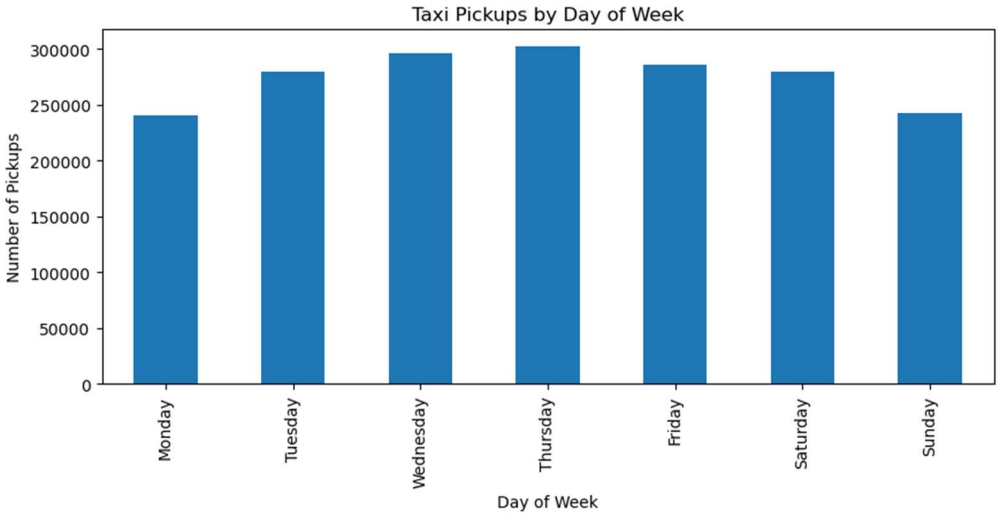


Fig.: Taxi pickups by days of the week

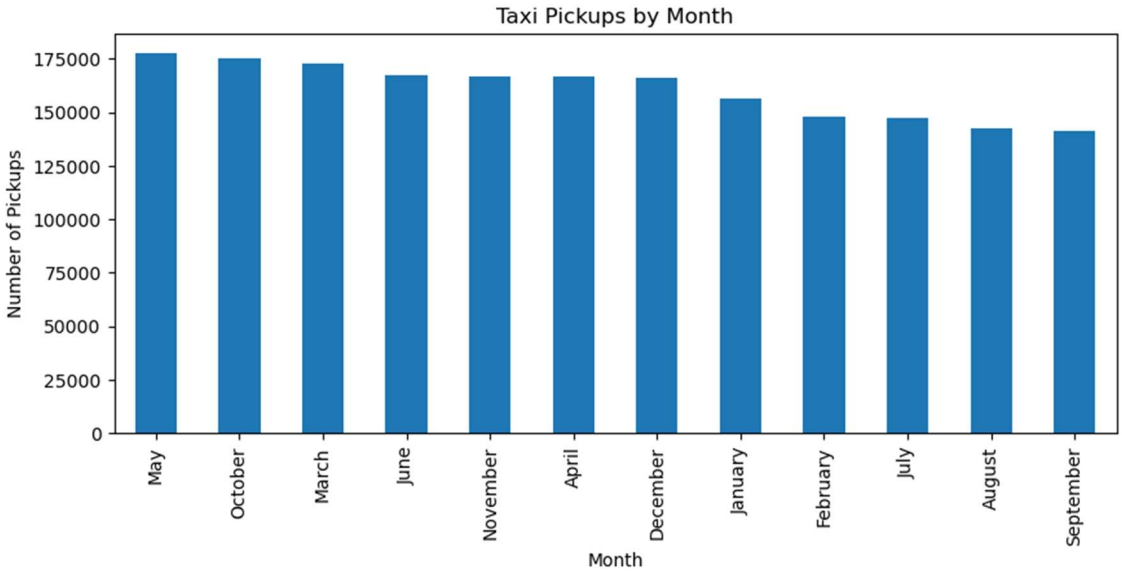


Fig.: Taxi pickups by month of year 2023

Findings:

- Peak demand occurs during **evening hours (5 PM – 7 PM)**.
- **Mid-week days (Wednesday and Thursday)** show the highest activity.
- **Summer and early winter months** exhibit relatively higher demand.

3.1.3. Filter out the zero/negative values in fares, distance and tips

Zero and negative values in:

- fare_amount - 527
- trip_distance - 24092
- tip_amount - 431773
- total_amount - 271

were analysed.

A filtered DataFrame excluding invalid values was created for financial analysis.

Zero-distance trips were retained only when pickup and dropoff zones were the same.

3.1.4. Analyse the monthly revenue trends

Monthly revenue was calculated by grouping data by month and summing total_amount.

Results showed:

- Higher revenue in mid-year and year-end months.
- Lower revenue in early months.

Quarterly analysis indicated that Q2 and Q4 contributed the highest revenue.

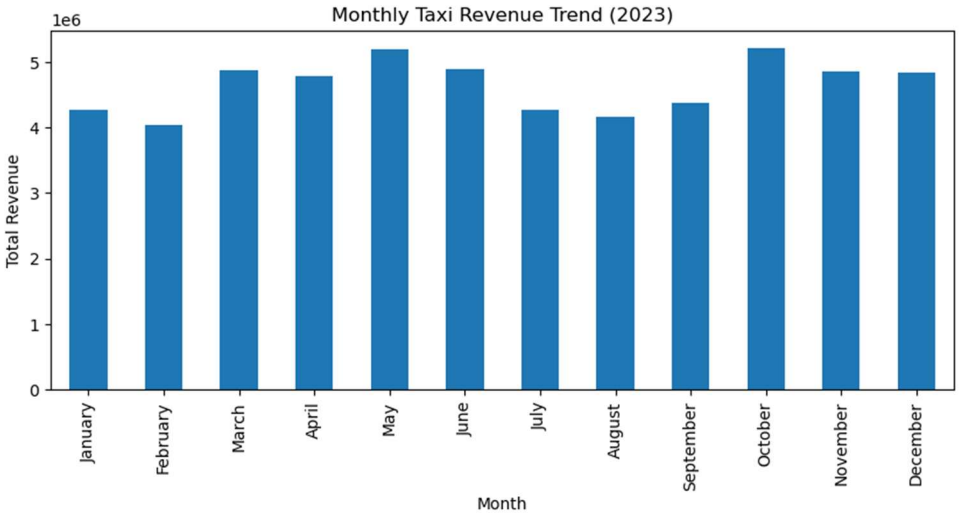


Fig.: Monthly taxi revenue trend (2023)

3.1.5. Find the proportion of each quarter's revenue in the yearly revenue

Trips with `trip_distance = 0` were excluded.

A scatter plot was created between `trip_distance` and `fare_amount`.

A strong positive correlation was observed, confirming that fare increases with distance.

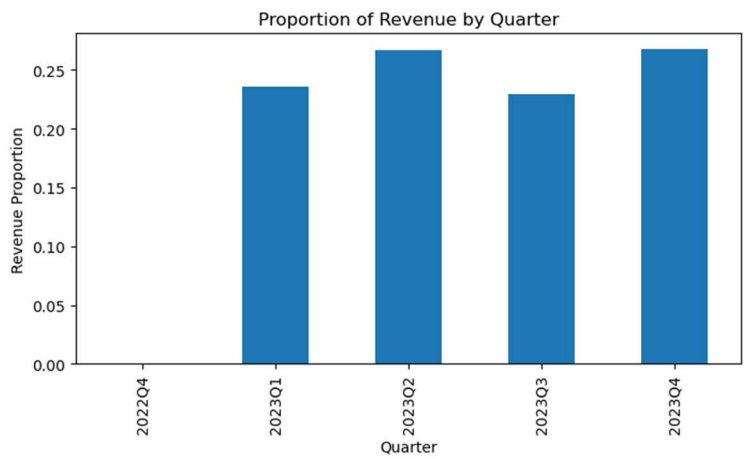


Fig.: Proportion of revenue by quarter

3.1.6. Analyse and visualise the relationship between distance and fare amount

Correlations were analysed between:

- Fare amount and trip distance - 0.1602449781553957

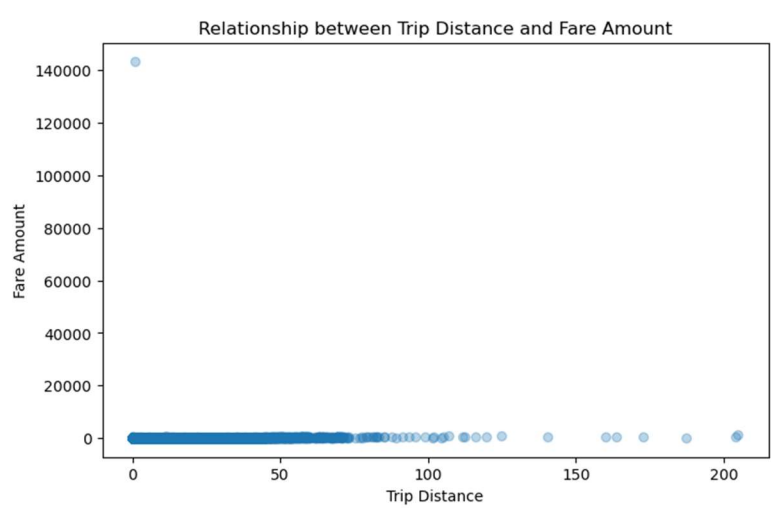


Fig.: Relationship between trip distance and fare amount

- Fare and trip duration - 0.045837699805851145

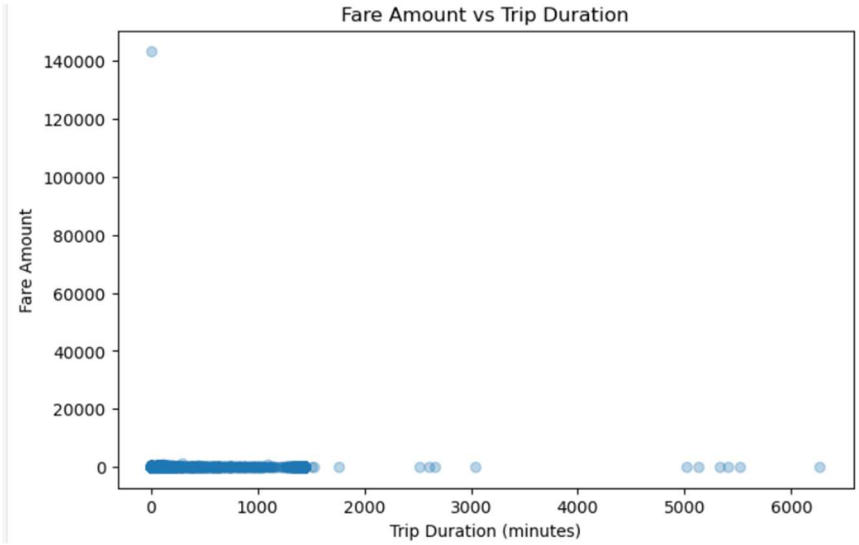


Fig.: Relationship between fare amount and trip duration

3.1.7. Analyse the relationship between fare/tips and trips/passengers

Fare and passenger count - 0.007461462258646217

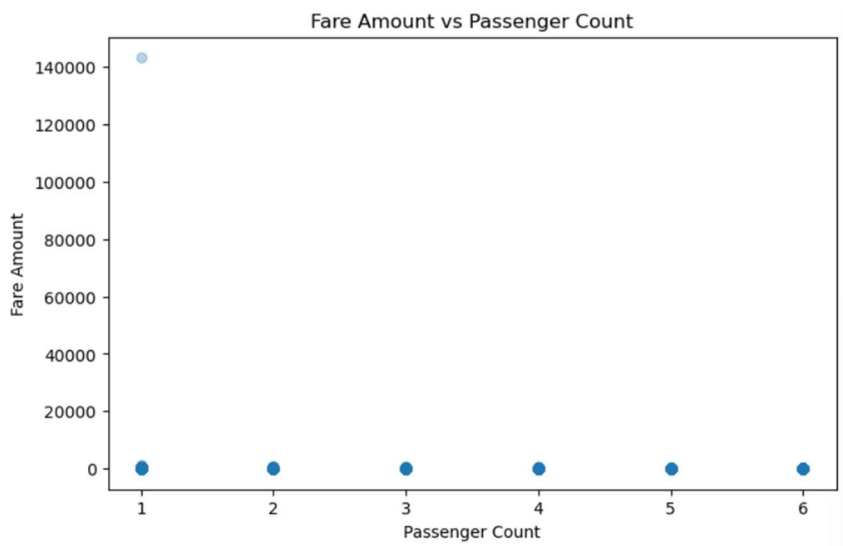


Fig.: Relationship between fare amount and passenger count

Tip and trip distance - 0.585655033403085

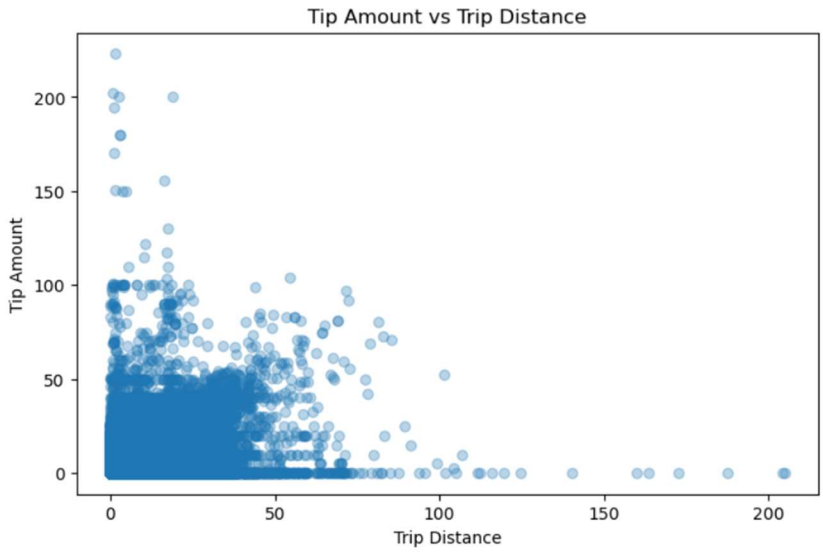


Fig.: Relationship between tip amount and trip distance

Results:

- Fare had strong correlation with duration.
- Passenger count had weak influence on fare.
- Tip amount moderately increased with distance.
-

3.1.8. Analyse the distribution of different payment types

Payment types were analysed:

1 – Credit card -1570596

2 – Cash - 333048

3 – No charge - 9451

4 – Dispute - 14405

- Credit card was most preferred.
- Cash was second.
- Disputes and no-charge cases were minimal.
- This shows high adoption of digital payments.

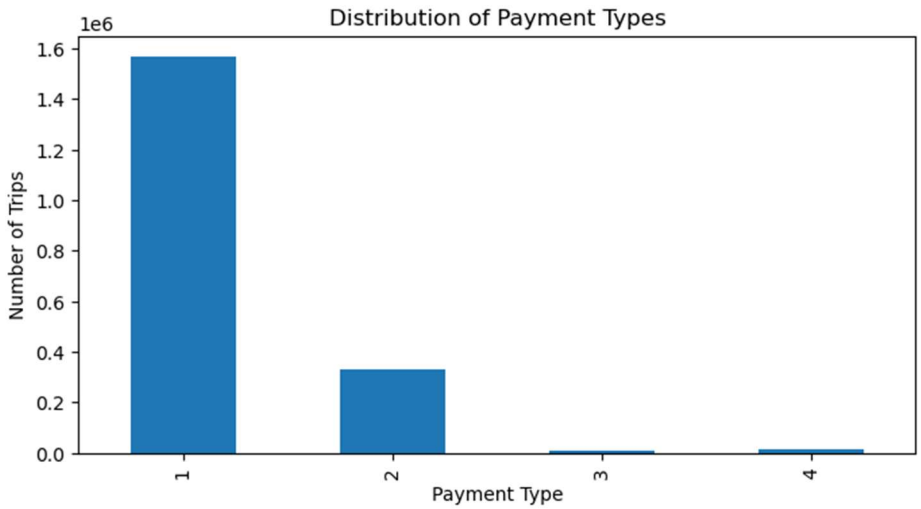
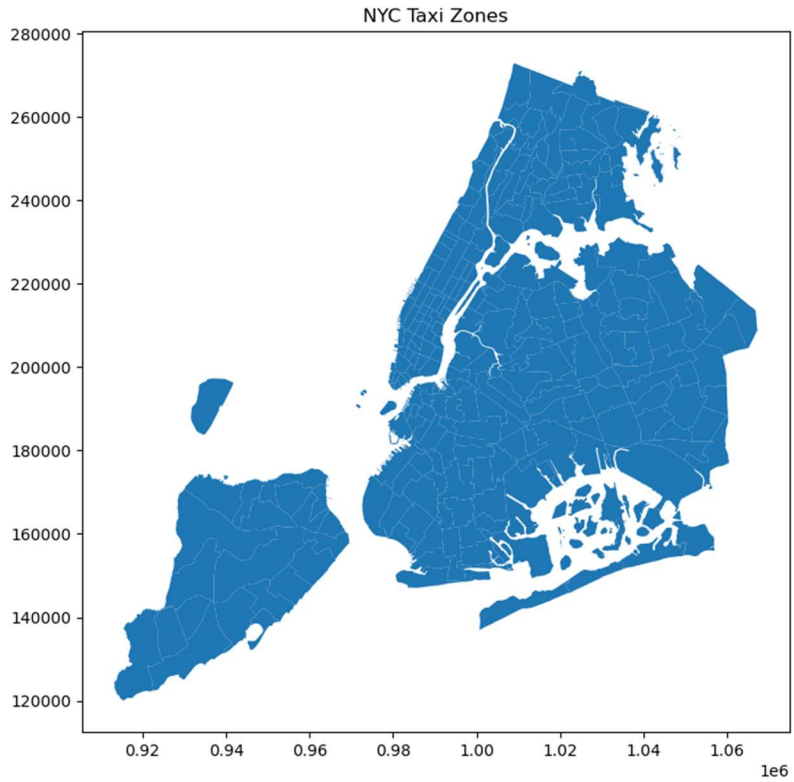


Fig.: Different types of payments

3.1.9. Load the taxi zones shapefile and display it

Taxi zone shapefile was loaded using GeoPandas.



3.1.10. Merge the zone data with trips data

The zones were merged with trip data using LocationID and PULocationID.

	PULocationID	PU_zone	PU_borough
0	138	LaGuardia Airport	Queens
1	161	Midtown Center	Manhattan
2	237	Upper East Side South	Manhattan
3	143	Lincoln Square West	Manhattan
4	246	West Chelsea/Hudson Yards	Manhattan

3.1.11. Find the number of trips for each zone/location ID

Group data by location IDs to find the total number of trips per location ID

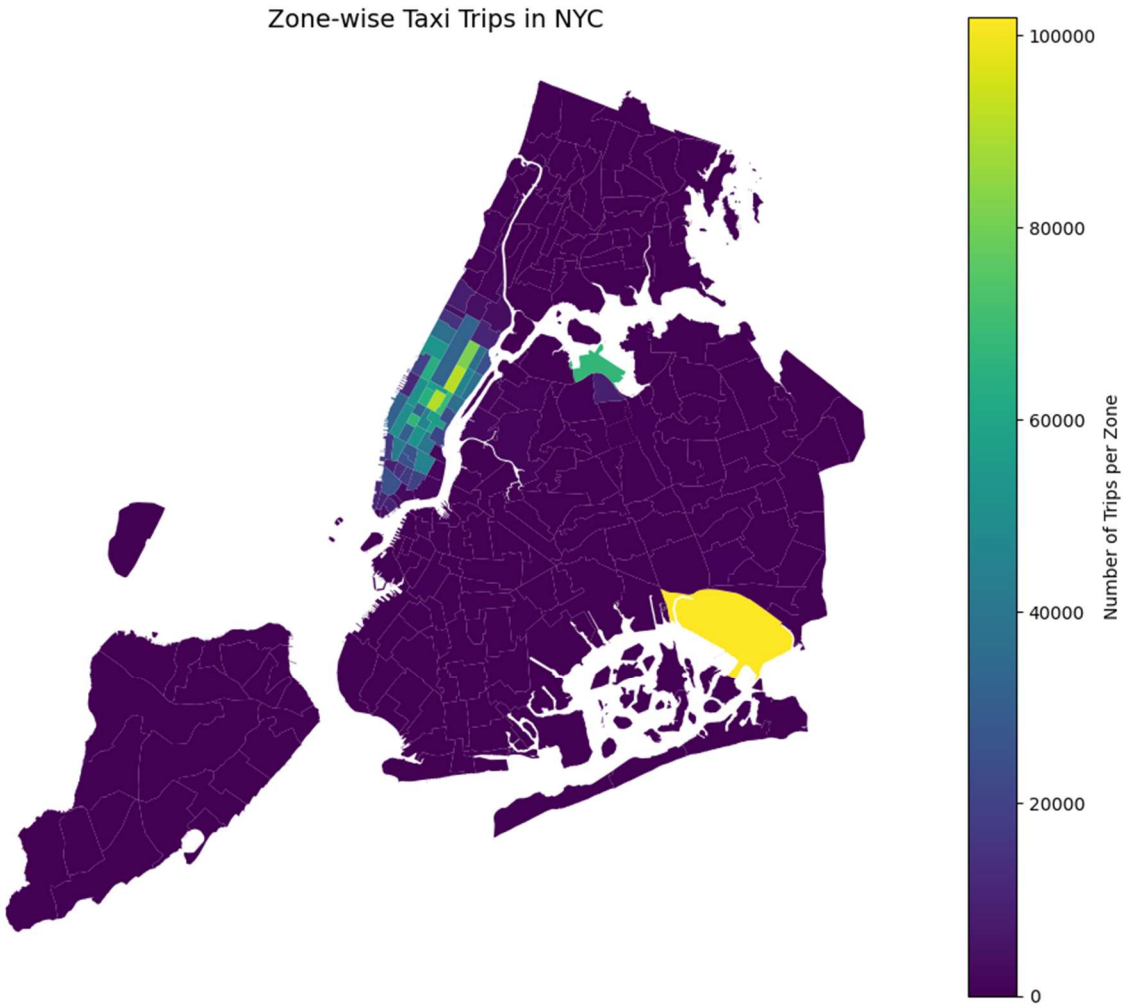
	PULocationID	total_trips	zone	borough
0	1	219	Newark Airport	EWR
1	2	2	Jamaica Bay	Queens
2	3	43	Allerton/Pelham Gardens	Bronx
3	4	1961	Alphabet City	Manhattan
4	5	14	Arden Heights	Staten Island

3.1.12. Add the number of trips for each zone to the zones dataframe

	OBJECTID	Shape_Leng	Shape_Area	zone_x	LocationID	borough_x	geometry	PULocationID	total_trips	zone_y	borough_y
0	1	0.116357	0.000782	Newark Airport	1	EWR	POLYGON ((933100.918 192536.086, 933091.011 19...	1.0	219.0	Newark Airport	EWR
1	2	0.433470	0.004866	Jamaica Bay	2	Queens	MULTIPOLYGON (((1033269.244 172126.008, 103343...	2.0	2.0	Jamaica Bay	Queens
2	3	0.084341	0.000314	Allerton/Pelham Gardens	3	Bronx	POLYGON ((1026308.77 256767.698, 1026495.593 2...	3.0	43.0	Allerton/Pelham Gardens	Bronx
3	4	0.043567	0.000112	Alphabet City	4	Manhattan	POLYGON ((992073.467 203714.076, 992068.667 20...	4.0	1961.0	Alphabet City	Manhattan
4	5	0.092146	0.000498	Arden Heights	5	Staten Island	POLYGON ((935843.31 144283.336, 936046.565 144...	5.0	14.0	Arden Heights	Staten Island

Table : Total trips added in the dataframe

3.1.13. Plot a map of the zones showing number of trips



3.1.14. Conclude with results

A choropleth map was plotted showing trip density.

Findings:

- Manhattan zones had highest trip volumes.
- Airport zones were highly active.
- Commercial and tourist areas showed dense traffic.
- Conclusion :
  Taxi demand peaks during evening hours and mid-week days.

- Revenue shows seasonal variation, with strong performance in Q2 and Q4.
- Fare is mainly dependent on trip distance and duration.
- Passenger count has minimal impact on fare.
- Tip amounts increase with trip distance.
- Credit cards dominate payment methods.
- Manhattan and airport zones are the busiest regions.

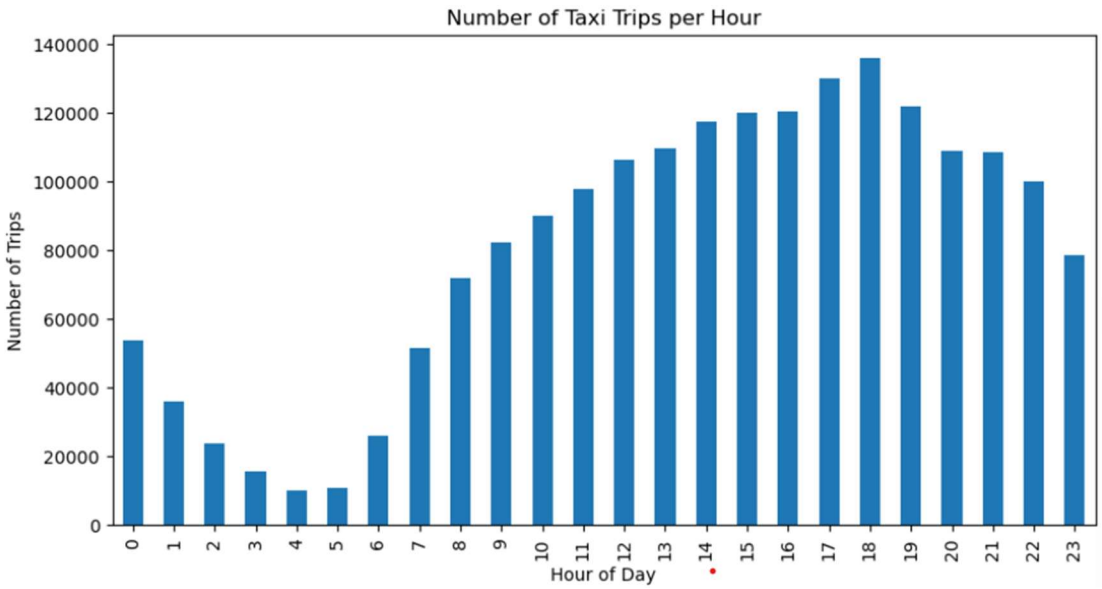
3.2. Detailed EDA: Insights and Strategies

3.2.1. Identify slow routes by comparing average speeds on different routes

Average speeds were calculated for routes at different hours.

Routes passing through central business districts showed lower speeds during peak hours, indicating congestion.

3.2.2. Calculate the hourly number of trips and identify the busy hours (18, 135945)



Top 5 pickup_hour

18 - 135945, 17 - 130071, 19 - 121989, 16 - 120303, 15 - 120286

3.2.3. Scale up the number of trips from above to find the actual number of trips

Trips per hour were calculated and scaled using the sampling fraction (0.05).

Actual trip counts were estimated as:

Actual Trips = Sampled Trips / 0.05

Top pickup and dropoff zones were identified.

Some zones acted mainly as origins, while others were destinations, such as airports and offices

3.2.4. Compare hourly traffic on weekdays and weekends

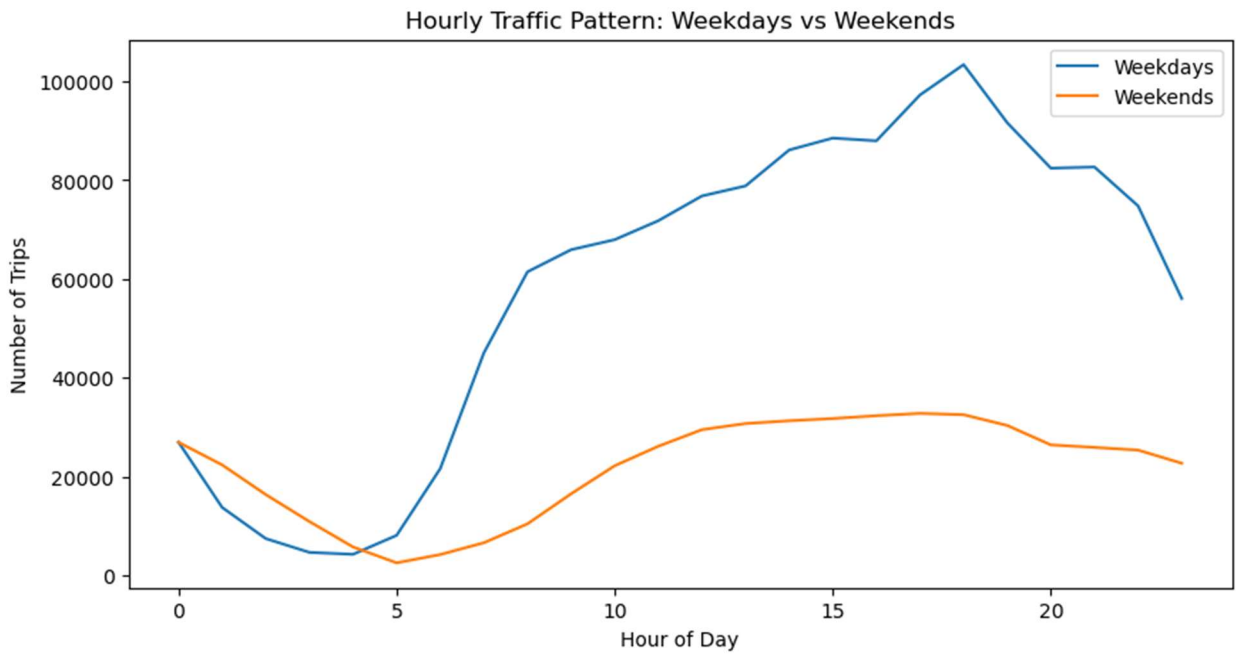


Fig.: Hourly traffic pattern on weekdays and weekends

- Weekdays show a clear commuter-driven pattern with a sharp rise in trips from 6 AM onwards.
- The highest demand on weekdays occurs between 5 PM and 7 PM, reflecting evening office return travel.
- Weekend traffic increases more gradually and peaks during the afternoon to early evening, indicating leisure-based travel.
- Overall, weekday trip volumes are consistently higher than weekends across all hours due to structured work schedules.

3.2.5. Identify the top 10 zones with high hourly pickups and drops

Top 10 pickup zones

PU_zone	
JFK Airport	101914
Upper East Side South	91474
Midtown Center	90469
Upper East Side North	81628
Midtown East	69161
LaGuardia Airport	67570
Penn Station/Madison Sq West	66821
Times Sq/Theatre District	64576
Lincoln Square East	64055
Murray Hill	57319

Top 10 drop off zones

DO_zone	
Upper East Side North	85557
Upper East Side South	81565
Midtown Center	75458
Times Sq/Theatre District	59442
Murray Hill	57053
Midtown East	54959
Lincoln Square East	54168
Upper West Side South	53940
Lenox Hill West	50973
East Chelsea	48757

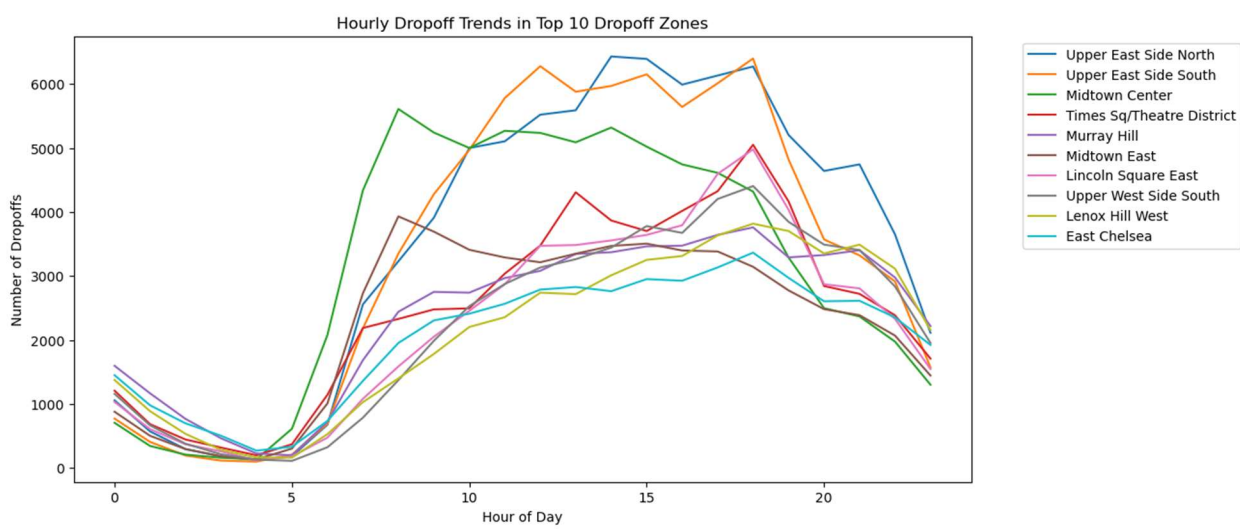


Fig.: Hourly dropoff trend

3.2.6. Find the ratio of pickups and dropoffs in each zone

```
Top 10 Highest Pickup/Dropoff Ratios
      pickups  dropoffs  pickup_drop_ratio
70     8785.0    1053.0           8.342830
132  102114.0   22105.0           4.619498
138   67682.0   23388.0           2.893877
186   66831.0   42280.0           1.580676
114   25421.0   18472.0           1.376191
43    32375.0   23584.0           1.372753
249   42553.0   32094.0           1.325886
162   69165.0   54959.0           1.258484
161   90476.0   75458.0           1.199025
100   31746.0   26652.0           1.191130

Top 10 Lowest Pickup/Dropoff Ratios
     pickups  dropoffs  pickup_drop_ratio
99       0.0       3.0           0.000000
30       0.0      18.0           0.000000
176      0.0      12.0           0.000000
245      0.0      31.0           0.000000
27       1.0      40.0           0.025000
221      1.0      35.0           0.028571
257     29.0     796.0           0.036432
1      219.0    5607.0           0.039058
115      1.0      23.0           0.043478
198     55.0    1050.0           0.052381
```

3.2.7. Identify the top zones with high traffic during night hours

	LocationID	Night_Pickups	zone
0	79	16377	East Village
1	132	15374	JFK Airport
2	249	13160	West Village
3	48	10982	Clinton East
4	148	10165	Lower East Side
5	114	9227	Greenwich Village South
6	230	8662	Times Sq/Theatre District
7	186	7336	Penn Station/Madison Sq West
8	164	6452	Midtown South
9	68	6349	East Chelsea

3.2.8. Find the revenue share for nighttime and daytime hours

Revenue share

Day	49065780.56
Night	6736515.13

Revenue percent

Day	87.927889
Night	12.072111

3.2.9. For the different passenger counts, find the average fare per mile per passenger

passenger_count	
1.0	10.985907
2.0	6.424208
3.0	3.928254
4.0	4.403484
5.0	1.702117
6.0	1.347235

- Trips with **1 passenger** have the **highest fare per mile per passenger (~10.99)**, as the entire fare is paid by a single rider.
- As the **passenger count increases**, the fare per mile per passenger **generally decreases**, indicating cost sharing among passengers.
- Trips with **5–6 passengers** show the **lowest fare per mile per passenger (≈1.35–1.70)**, making them the most economical on a per-person basis.
- This trend highlights the advantage of **shared rides**, which reduce individual travel costs while maintaining overall fare revenue.

3.2.10. Find the average fare per mile by hours of the day and by days of the week

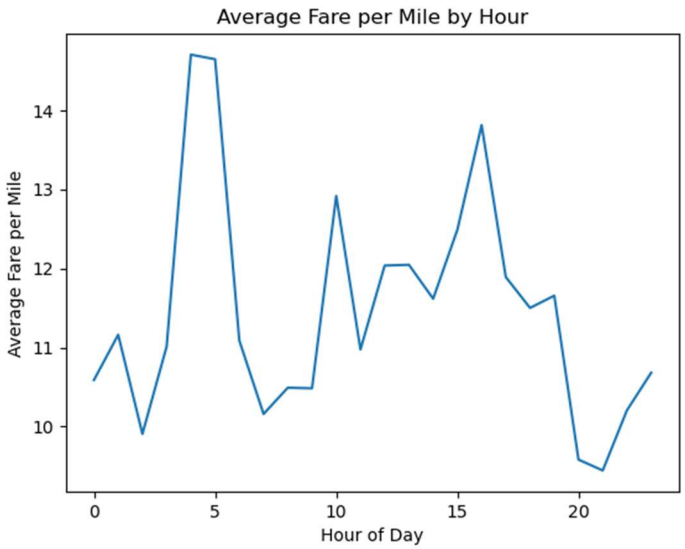


Fig.: average fare per mile by hour

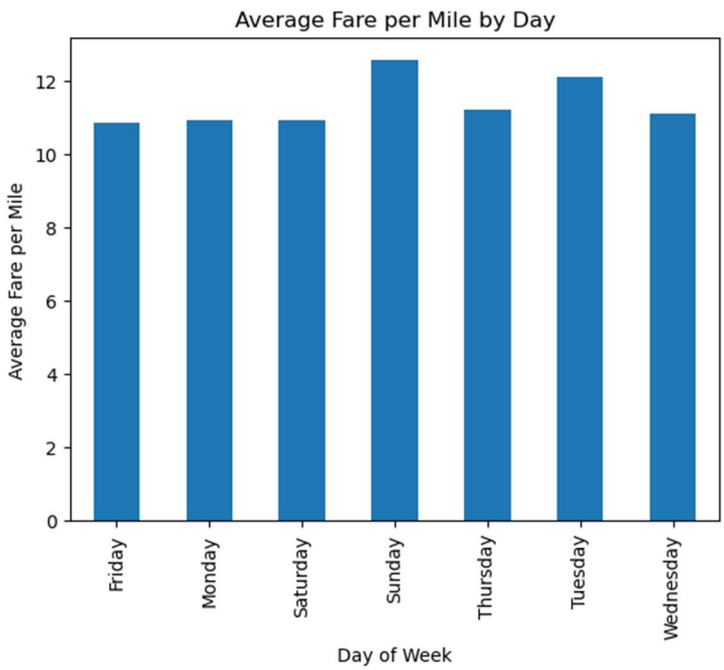
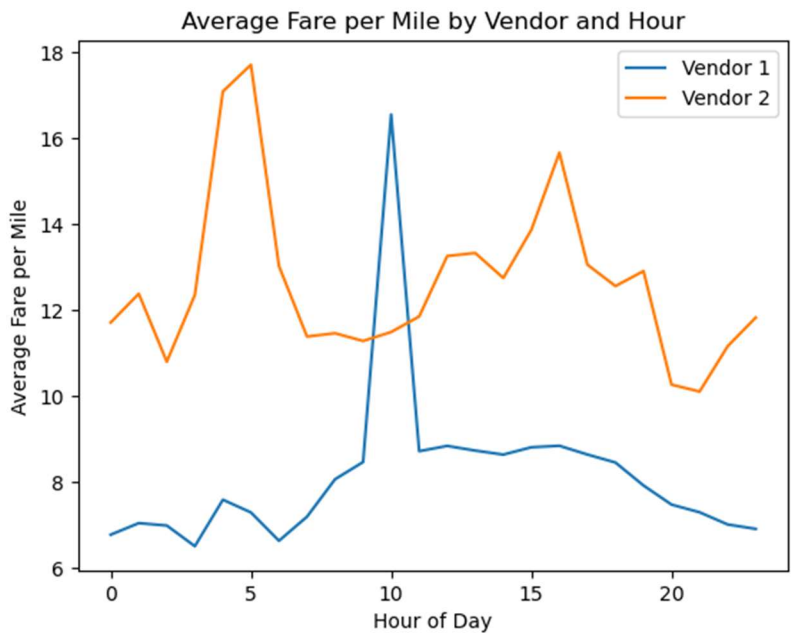


Fig.: average fare per mile by days of the week

- The **average fare per mile varies across hours**, with **early morning hours (4–6 AM)** showing the **highest values (~14–15)**, likely due to lower cab availability and night-time surcharges.

- During **daytime and evening hours**, fare per mile remains relatively stable, generally ranging between **10–12**, with a slight increase during **evening peak hours** due to higher demand and congestion.
- Day-wise analysis shows that **Sunday and Tuesday record the highest average fare per mile**, while **Friday and Saturday have comparatively lower values**, indicating higher demand-driven pricing on certain weekdays and premium fares during low-supply periods.

3.2.11. Analyse the average fare per mile for the different vendors



- **Vendor 2** consistently charges a **higher average fare per mile** across most hours compared to Vendor 1.
- The difference between vendors is most pronounced during **early morning and late evening hours**, indicating possible premium pricing by Vendor 2 during low-supply periods.
- **Vendor 1** shows relatively **stable and lower fare per mile values**, suggesting a more uniform pricing strategy across the day.

3.2.12. Compare the fare rates of different vendors in a distance-tiered fashion

	VendorID	distance_tier	fare_per_mile
0	1	0-2 miles	10.634455
1	1	2-5 miles	6.378955
2	1	5+ miles	4.422504
3	2	0-2 miles	17.903077
4	2	2-5 miles	6.548410
5	2	5+ miles	4.502418

3.2.13. Analyse the tip percentages

```
distance_group
0-2     22.415844
2-5     18.757743
5-10    19.398931
10+     15.789250
```

```
passenger_count
1.0    20.917382
2.0    19.725986
3.0    19.024114
4.0    17.445196
5.0    20.500106
6.0    20.555222
```

```
pickup_hour
0     20.508077
1     20.625301
2     20.623704
3     20.459764
4     18.286375
5     17.467478
6     18.442166
7     19.571571
8     20.029041
9     19.717923
10    19.250896
11    19.168779
12    19.120282
13    19.054640
14    19.032831
15    18.998561
16    20.914962
17    24.360268
18    22.223355
19    22.181292
20    21.496390
21    21.511269
22    21.305280
23    20.675190
```

Table: Analysis of tip percentages by distance, passenger count and pickup hour

3.2.14. Analyse the trends in passenger count

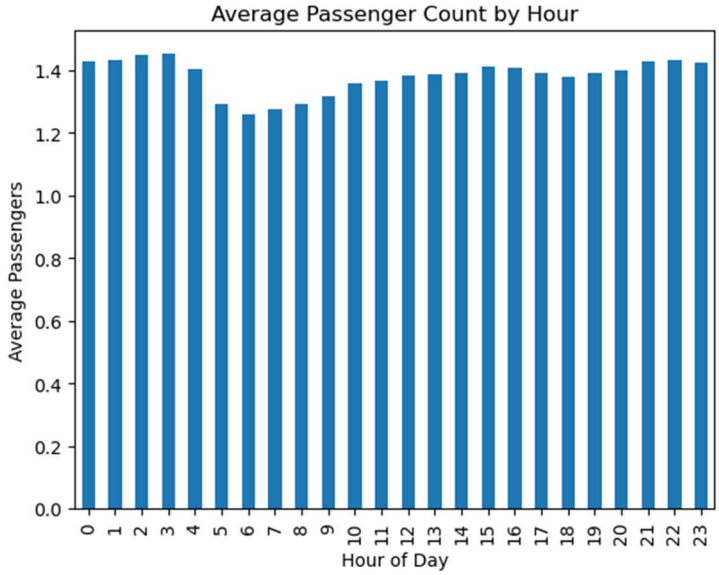


Fig.: Average passenger count by hour

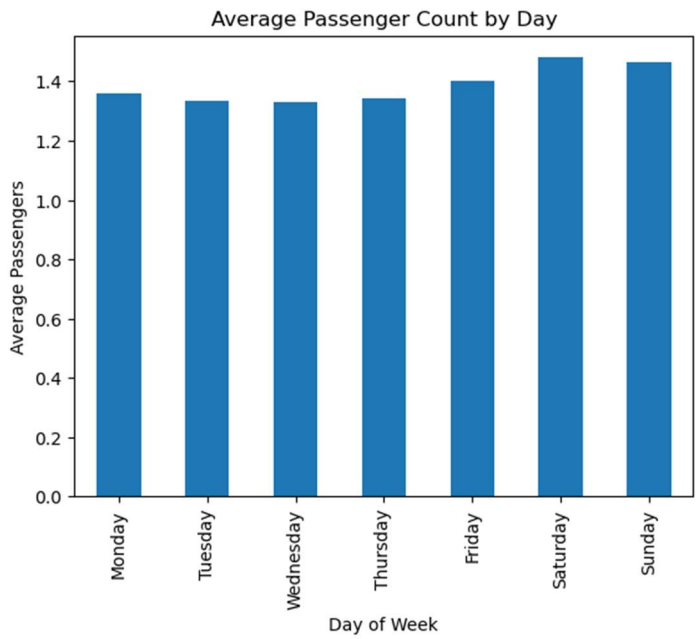
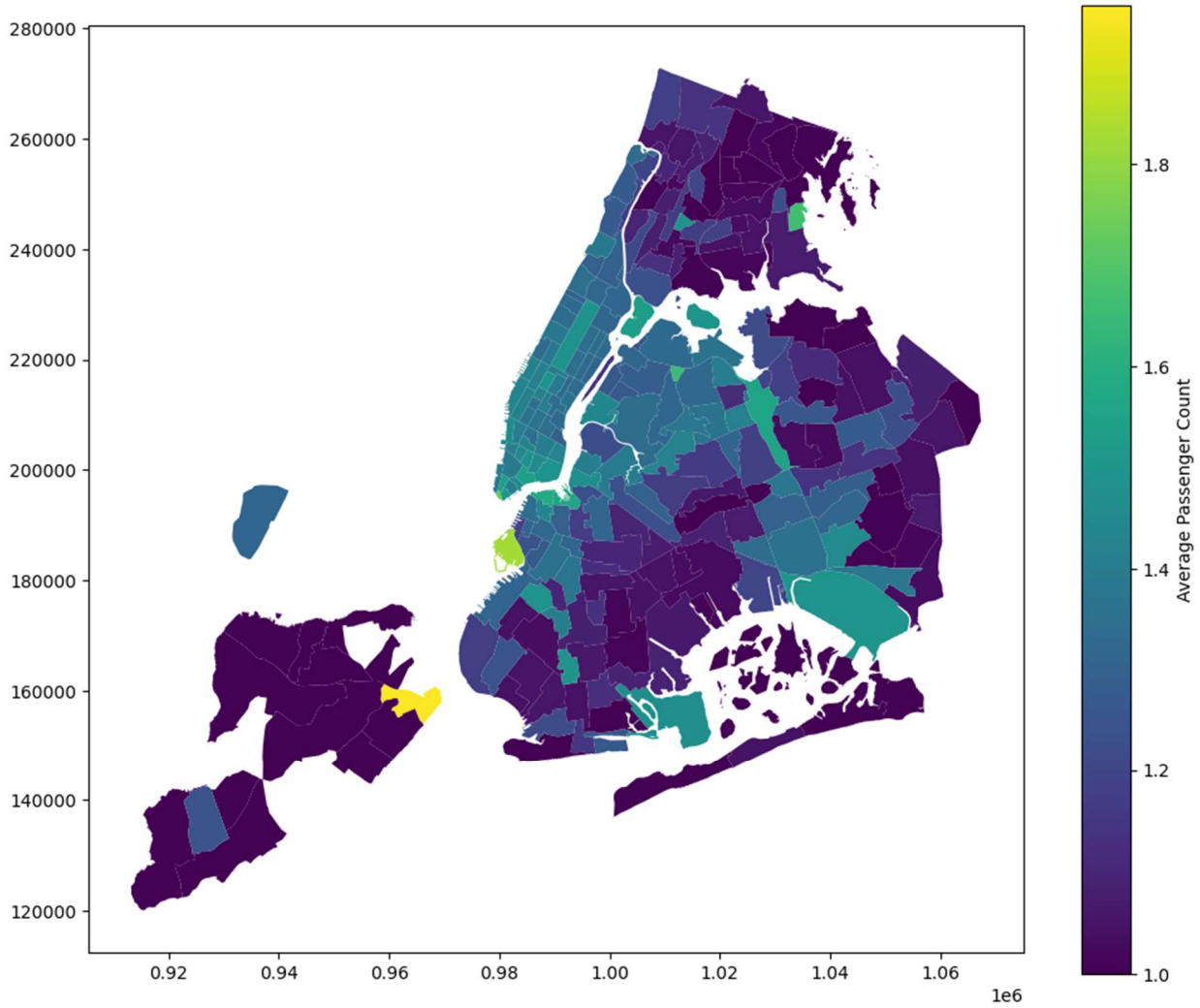


Fig.: Average passenger count by day

- The **average passenger count remains fairly stable across all hours**, ranging between **1.3 and 1.45**, indicating that most trips involve **single or two passengers**.
- A slight increase in passenger count is observed during **late night and early morning hours**, possibly due to shared rides after social or leisure activities.

- Day-wise analysis shows that **weekends (Saturday and Sunday)** have a **higher average passenger count** compared to weekdays, suggesting increased group travel during leisure days.

3.2.15. Analyse the variation of passenger counts across zones



- The average passenger count across most zones lies between 1 and 1.5, indicating that single-passenger trips dominate citywide.
- Higher average passenger counts are observed in airport zones and select commercial areas, suggesting group travel for long-distance or shared trips.
- Residential and outer borough zones generally show lower average passenger counts, reflecting routine, individual travel patterns.

3.2.16. Analyse the pickup/dropoff zones or times when extra charges are applied more frequently.

Night extra charges - 96.24453643635587

Day extra charges - 62.30747051430654

4. Conclusions

4.1. Final Insights and Recommendations

4.1.1. Recommendations to optimize routing and dispatching based on demand patterns and operational inefficiencies.

Peak demand happens on the weekends, late at night, and during office hours, according to an analysis of hourly and zonal demand.

Due to congestion, some routes and zone pairs were found to be slow-moving, particularly during peak hours.

In order to maximize dispatching and routing:

- Real-time traffic and historical congestion data should be used to dynamically route taxis.
- During peak hours, more cars ought to be placed on slow-moving, high-demand routes.
- Drivers should be advised to take alternate routes in order to avoid traffic jams.
- To cut down on passenger wait times, dispatch systems ought to give priority to assigning nearby taxis.
- It is important to keep an eye on and provide sufficient taxi availability for nighttime routes with steady demand.

By taking these actions, you can shorten travel times, increase the number of trips per cab, and improve fuel efficiency.

4.1.2. Suggestions on strategically positioning cabs across different zones to make best use of insights uncovered by analysing trip trends across time, days and months.

According to geographic analysis, the most pickups and drop-offs occur in major residential areas, airport zones, and central business districts.
On weekdays, demand is greatest in commercial districts, and on weekends and evenings, it is highest in entertainment districts.

Considering these patterns:

- During peak hours, more taxis ought to be placed in areas with high demand, and during morning and evening commutes, they ought to be moved to residential areas.
- There should always be an adequate supply of taxis at airports and transit hub areas.
- More taxis ought to be stationed close to shopping centers, eateries, and entertainment venues on the weekends and late at night.
- Fleet deployment planning should take monthly and seasonal trends into account.
- Forecasts of demand should guide the monitoring and supply of underutilized zones.

This minimizes idle time and guarantees balanced cab availability.

4.1.3. Propose data-driven adjustments to the pricing strategy to maximize revenue while maintaining competitive rates with other vendors.

Variations across time, vendors, and passenger counts are revealed by fare per mile analysis.

In high-congestion areas, at night, and during peak hours, fares are higher.

Customers are more likely to tip for faster and smoother rides, according to tip analysis.

The following pricing strategies are suggested in light of these insights:

- Instead of using abrupt spikes in prices during periods of high demand and peak hours, use moderate dynamic pricing.
- To boost the number of trips, provide reduced prices during off-peak hours.
- To promote ride sharing, implement bundled pricing for trips with multiple passengers.
- In order to preserve service availability, offer incentives to drivers in areas with high demand.
- Keep an eye on rivals' prices to make sure rates stay affordable while optimizing profits.

These tactics increase profitability while preserving customer satisfaction.